



# Evaluation of structural, elastic properties and nuclear radiation shielding competence of Nd<sup>3+</sup> doped lithium-zinc-phosphate glasses

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## ABSTRACT

Nd<sup>3+</sup> doped lithium-zinc-phosphate glasses were synthesized by melt-quenching technique. FTIR spectra were recorded in the wavenumber range of 400–4000 cm<sup>-1</sup>. Elastic parameters such as Debye temperature and fractal bond connectivity were calculated. The results revealed that these parameters were influenced by the glass system's composition and dopant concentration. The mass and linear attenuation coefficients (MAC, LAC) of gamma-ray photons at the energy range from 0.015 to 15 MeV have been investigated using the Phy-X/PSD software program as well as the XCOM database program. Both MAC and LAC increase with the increase in the amount of Nd<sub>2</sub>O<sub>3</sub>. Also, the mean free path values and the half-value layer decrease while the density increases with the increase of the Nd<sub>2</sub>O<sub>3</sub> content. Moreover, the fast neutron removal cross section was also presented and the results were compared with the concrete and commercial shielding glasses.

## 1. Introduction

Evaluation of various gamma-ray factors such as mass attenuation coefficient (MAC) and half-value layer (HVL) is necessary for the practical use of shielding materials in the domains of radiation protection, dosimetry, and nuclear medicine diagnostics. At radiation locations, particularly in jobs near nuclear reactors, common materials such as lead-containing products and concrete are now utilized to shield employees from dangerous X-rays, gamma-rays, and neutrons. However, in the field of radiation shielding materials, the environmental toxicity of Pb and its dangerous nature to human health are major issues for researchers who want to replace Pb with other high Z-materials such as Nd, Bi, Gd, and Ba [1–3]. Moreover, concrete is opaque to transmit visible light. Concrete has many disadvantages such as cracks, which formed due to the water content evaporation process with the high-energies of gamma-rays. Furthermore, glass systems are suitable alternatives to concrete as efficient gamma-ray shielding materials due to their transparency to visible light, ease of production, and the ability to quickly modify their physical and optical characteristics through composition and preparation processes. As a result, several types of

glasses, such as borate, phosphate, tellurite, silicate, borophosphate, borosilicate and borotellurite glasses have been studied for their radiation shielding properties by various researchers [2,4–10]. According to these works, the composition of glass has a significant influence on its shielding and other properties. In addition, these glasses have been studied after adding some modifiers in order to change their structures for various applications. For example, phosphate glasses containing Li as modifiers can be used in batteries [11,12]. Also, Phosphate glasses containing Zn atoms have varieties of applications such as photovoltaic cells, the vitrification of radioactive waste, thermoelectric sensors, and the enhancement of the band gap. The addition of Zinc atoms enhances the chemical stability and durability of phosphate glasses by substituting bridging P-O-P bonds with more moisture-resistant P-O-Zn bonds. Chemical durability can be improved by adding ZnO and it leads to the decrease of the glass transition temperature and to donate a broad forming region of glass [13,14]. Studies of ultrasonic characteristics as a function of the host matrix give important information on atomic and molecular configurations in the glass. The dependence of mechanical and optical properties on the glass composition of soda-lime-silica (SL) glasses has been studied [15–17]. The reported results of these studies

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confirmed that the mechanical and optical properties of SLS glasses enhanced through peanut shell and eggshell powder additions in the glass structure. Glasses doped with  $\text{Nd}^{3+}$  ions are desired for their chemical resistance, hardness, and their applications in laser and optoelectronics technology [18,19]. Therefore, the relationship between gamma-rays shielding glass materials and their mechanical properties is very important. The elastic characteristics of material define its mechanical behavior, and their study is critical for understanding and interpreting the nature of bonding in solid-state systems.

In this work, the structural, mechanical, and nuclear radiation shielding (NRS) properties of lithium-zinc-phosphate (LZP) glass matrix doped with various concentrations of  $\text{Nd}^{3+}$  ions were reported. The examination of elastic characteristics is required to determine the glasses' suitability for the target technical applications, particularly fiber drawing. Therefore, our research focused on the elastic and shielding characteristics of these glasses. Our study covered the elastic moduli such as Poisson ratio, softening temperature, and Debye temperature and NRS properties such as the MAC, the HVL, and the electron density ( $N_{\text{eff}}$ ). In addition to providing a brief overview of the nature of the bonding properties of the glass material under consideration, as well as knowing its internal structure via FTIR and the impact of neodymium oxide additions on the aforementioned properties.

## 2. Experimental procedures

Lithium-zinc-phosphate glass was used as a host matrix for  $\text{Nd}^{3+}$  ions. The prepared glass system is  $(100-x)(0.2 \text{ mol\% Li}_2\text{O}-0.4 \text{ mol ZnO}-0.4 \text{ mol\% P}_2\text{O}_5):x\text{Nd}_2\text{O}_3$  (referred as Nd0, Nd2, Nd4, Nd6 and Nd8, for  $x = 0, 2, 4, 6$  and  $8 \text{ wt\%}$ , respectively). High purity raw materials suitable for optical glasses were used. Amounts of analar  $\text{Li}_2\text{CO}_3$ ,  $\text{ZnO}$ , and  $\text{P}_2\text{O}_5$  were weighed and doped with  $x$  weight percent  $\text{Nd}_2\text{O}_3$  from Laboratory reagent grades. These chemicals were combined and placed in alumina crucibles, which were then placed in an electric furnace, with an open environment, and kept at  $300^\circ\text{C}$  for 1 h. This process of heat treatment allows the  $\text{P}_2\text{O}_5$  to decompose and react with  $\text{Li}_2\text{CO}_3$ ,  $\text{ZnO}$  and  $\text{Nd}_2\text{O}_3$  before melting. The combinations were then transferred to a second furnace for 1 h at  $1150^\circ\text{C}$ . To guarantee homogeneity, the melts were sometimes agitated with an alumina rod. The melt was cast into two mild-steel split moulds heated to  $250^\circ\text{C}$  to form glass rods 1.3 cm long and 1 cm in diameter. Then, it was immediately moved to an annealing furnace and annealed for 1 h at  $300^\circ\text{C}$ . The furnace was then turned off, and the glass was allowed to cool to room temperature gradually.

An ultrasonic flaw detector, USM3, and a 3055-time base oscilloscope were used to detect ultrasonic velocity in the glass sample. By matching transducers, the pulse section creates electrical pulses that are translated into ultrasonic signals. The ultrasonic pulse passes through the sample coupled to the transducer, registering an echo and returning to the transducer each time. Due to attenuation in the sample, the amplitude of subsequent echoes decreases experimentally, and numerous echoes appear on the screen. The following equation may be used to determine the ultrasonic wave velocity:

$$v = 2 \times \text{sample thickness}(d)/\text{time interval}(t) \quad (1)$$

At room temperature, all velocity measurements were performed at a nominal frequency of 5 MHz. The ultrasonic probe's frequency response covers the range of 3.6 to 5.4 MHz. The measurements were performed three times to ensure that the data was consistent.

JASCO FT/IR-460 plus (Japan) spectrometer with KBr disc technique was used to record the FTIR spectra of samples within the wavenumber range of  $4000\text{--}400 \text{ cm}^{-1}$ . The mass attenuation coefficient (MAC) of gamma-ray photons at the energy range from 0.015 to 15 MeV were simulated using the Phy-X/PSD software program as well as the XCOM database program.

## 3. Results and discussion

### 3.1. Fourier transformed infrared (FTIR) spectra

The FTIR absorbance spectra were recorded in the wavenumber range from  $4000$  to  $400 \text{ cm}^{-1}$  for all samples as shown in Fig. 1 and their assignments reveal the following bands:

- i- The absorption peak observed at around  $523\text{--}541 \text{ cm}^{-1}$  is assigned to the Nd-O bond stretching vibrations [20].
- ii- The absorption band at  $\sim 753 \text{ cm}^{-1}$  is attributed to the P-O-P symmetric stretching vibrations,  $\nu_s(\text{P-O-P})$  [21].
- iii- The located band at around  $920\text{--}927 \text{ cm}^{-1}$  is assigned to the asymmetric stretching vibrations of P-O-P,  $\nu_{\text{as}}(\text{P-O-P})$  [22].
- iv- The absorption bands at wavenumber in the region between  $990$  and  $1000 \text{ cm}^{-1}$  are related to the symmetrical stretching vibration of P-O-P bridging oxygen of the metaphosphate group [23].
- v- The absorption band at around  $1106 \text{ cm}^{-1}$  may be due to the symmetric stretching mode of phosphorus and non-bridging oxygen on  $\text{Q}^2$  groups [24].
- vi- The absorption peak in the region between  $1250$  and  $1262 \text{ cm}^{-1}$  is attributed to the stretching vibrations of  $\text{P=O}$  [25]. This band is superposed with both the asymmetric stretching vibrations of the two non-bridging oxygen atoms bonded to a phosphorus atom in a  $\text{Q}^2$  phosphate tetrahedron,  $\nu_{\text{as}}(\text{PO}_2)$  [26].
- vii- The asymmetric stretching vibrations of  $(\text{PO}_2)^-$  in the  $(\text{Q}^2)$  units are related to the band at around  $1300\text{--}1317 \text{ cm}^{-1}$  [27].
- viii- The absorption bands detected at  $1625\text{--}1650 \text{ cm}^{-1}$  are related to the stretching and bending vibrations of P-O-H bridges, as well as  $\text{H}_2\text{O}$ , which is related to moisture absorption during KBr pellet preparation [28].
- ix- The small absorption peak appeared at  $2214 \text{ cm}^{-1}$  for the undoped glass is related to  $2\nu_3 + \nu_1$  and  $\nu_4$  normal vibrations of  $\text{PO}_4^{3-}$  ion [29]. This peak dissolves for the glasses doped with  $\text{Nd}_2\text{O}_3$ .
- x- The antisymmetric stretching of the adsorbed water molecules is responsible for the absorption band at  $2358 \text{ cm}^{-1}$  [28].
- xi- The existence of hydrogen bonds [28] and stretching modes of O-H bonds results in double weak absorption bands in the regions  $2853\text{--}2858 \text{ cm}^{-1}$  and  $2919\text{--}2925 \text{ cm}^{-1}$ . The intensity of these bands decreases with the increase of the  $\text{Nd}_2\text{O}_3$  content leading to

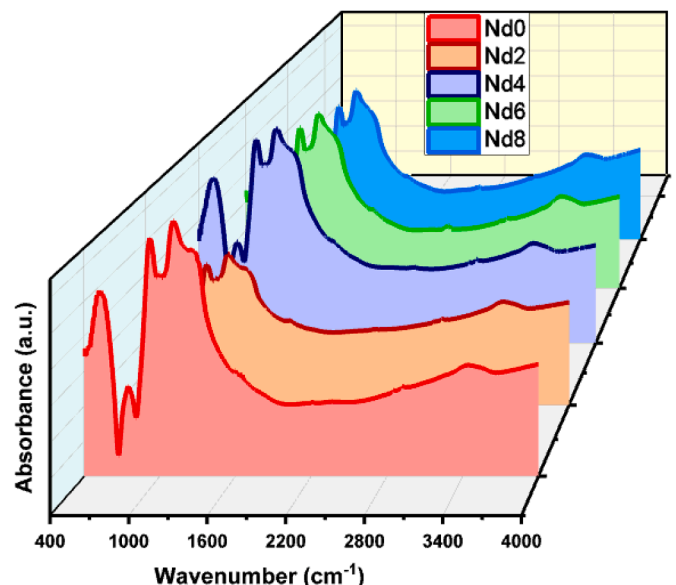


Fig. 1. FTIR absorption spectra of glass samples.

the enhancement of the lower non-radiative losses within the current glass system.

- xii- The O-H stretching vibrations of the hydroxyl group and molecular water are responsible for the wideband seen in the range 3433–3450  $\text{cm}^{-1}$  [28]. The intensities of these bands decrease with the increase of the  $\text{Nd}_2\text{O}_3$  content that indicating the decrease of the hygroscopic nature of the glasses.

The deconvoluted spectra of the Nd0 and Nd8 glass samples are depicted in Fig. 2. Table 1 shows the variations of the deconvolution parameters (the band center,  $x_c$ , area,  $A$ , and full width at half maximum,  $w$ ) with the  $\text{Nd}_2\text{O}_3$  addition. Based on deconvoluted data, the general behavior corresponding to the  $\text{PO}_4^{3-}$  ionic tetrahedron is shifted to lower wavenumber, and based on Table 1, it can be observed that the symmetric stretching bonds of P-O-P and ionic vibration bonds of  $\text{PO}_4^{3-}$  displayed a reverse behavior for the variations of their wavenumber positions with  $\text{Nd}_2\text{O}_3$  content. Whereas the  $\text{PO}_4^{3-}$  mode has weakened, the P-O-P mode has strengthened as the  $\text{Nd}_2\text{O}_3$  content is increased. It can also be observed that both the symmetric vibrations of P-O-P and/or Zn-O-Zn groups and the vibrations of  $\text{P-O}^-$  groups are displayed a reverse behavior for the variations of their wavenumber positions with  $\text{Nd}_2\text{O}_3$  content. The appearance this reverse behavior in various composition regions may be due to the emergence of structural alterations as a result of the various characteristics of the added cations to the phosphate matrix,

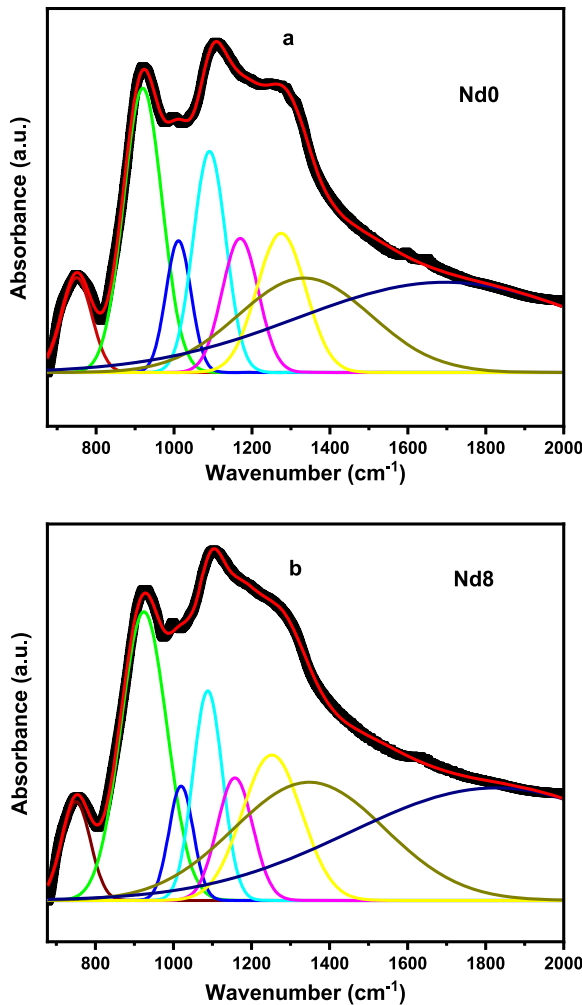


Fig. 2. The deconvolution data of the FTIR absorption spectra of: (a) Nd0 and (b) Nd8 samples.

i.e., the way in which the additional cations interact with the phosphate matrix. Interstitially or substitutionally, an additional cation might enter the glass network.

### 3.2. Bonding properties

Ionic-covalent and ignored metallic glass oxides with free electrons or electronic imperfections are important for electrolyte applications. The ionic characteristic factor (ionicity)  $I_c$  of glass, was calculated using the electronegativity ( $\Delta\chi$ ) values of the various oxides in the glass composition. The  $I_c$  is a term used to characterize the nature of glass bonding and is given by [30]:

$$I_c\% = \left[1 - e^{-0.25(\Delta\chi)^2}\right] \times 100 \quad (2)$$

Table 2 shows that the  $I_c\%$  values of all glass samples approach fifty percent, confirming the ionic character of the current glass.

### 3.3. Elastic properties

The elastic moduli of prepared glasses give information about the stress-strain connection in the glass system in great detail. The same characteristic is also used to investigate the strength of a material when it is subjected to force. Additionally, the Elastic moduli are another important factor to consider when choosing glass materials for a certain application. Furthermore, because the elastic parameter of glass materials is directly related to the interactions between atoms, it provides important information regarding the structure of these amorphous solids. The variation in the nature of the chemical bond and its strength in the glass structure is usually incorporated into Young's modulus which has the ability to determine the fracture behavior involved in the glass. On the other hand, the bulk modulus is more sensitive in exploring changes in the cross-link density and the bond stretching force constant, as reported in [31]. The insertion of the rare earth ion into the glass matrix assists in the reduction of inter-atomic spacing or bond length between the atoms, which may be due to an increase in the stretching force constant of the bonds within the glass matrix.

The longitudinal ( $V_L$ ) and shear ( $V_S$ ) ultrasonic velocities increase with increasing  $\text{Nd}_2\text{O}_3$  concentration, as seen in Fig. 3. The longitudinal velocity increased from 5145 to 5280  $\text{ms}^{-1}$ , and the shear velocity increased from 2805 to 3096  $\text{ms}^{-1}$ . The rise in ultrasonic velocities shows that the glass network is strengthening and the samples are becoming more rigid [32]. The increment in the ultrasonic velocities  $V_L$  and  $V_S$  with  $\text{Nd}_2\text{O}_3$  concentration is due to the substitution of low-strength P-O-P chemical bonds with high-strength P-O-Nd or Zn-O-Nd chemical bonds. Furthermore, due to the tiny difference in Pauling ionic radii,  $\text{Nd}^{3+}$  ions may substitute  $\text{Li}^+$  in the LZP glass network, so modifying the network compacts the glass structure and improves the network's rigidity, and rises ultrasonic velocities [33].

Based on the ultrasonic velocities and densities of samples, the longitudinal (L), shear (G), bulk (K), Young's modulus (E), Poisson's ratio ( $\sigma$ ) and mean sound velocity ( $U_m$ ) of the glass samples were calculated using the following equations [34].

$$\text{Longitudinal elastic modulus, } L = \rho V_L^2 \quad (3)$$

$$\text{Shear elastic modulus, } G = \rho V_S^2 \quad (4)$$

$$\text{Bulk elastic modulus, } K = \frac{3L - 4G}{3} \quad (5)$$

$$\text{Young's elastic modulus, } E = 2G(1 + \sigma) \quad (6)$$

$$\text{Poisson's ratio, } \sigma = \frac{(L - 2G)}{(2L - 2G)} \quad (7)$$

**Table 1**

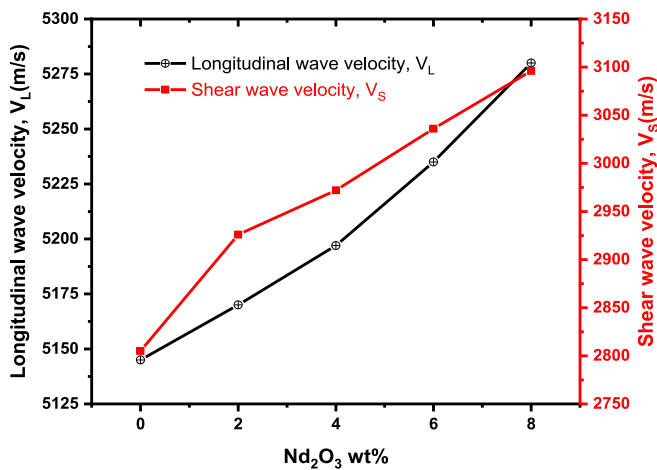
Deconvolution parameter of the infrared spectra of samples.

| Sample code |       | Peak position( $X_c$ ), area(A) and full width at half maximum (w) |       |        |        |        |        |        |        |
|-------------|-------|--------------------------------------------------------------------|-------|--------|--------|--------|--------|--------|--------|
| Nd0         | $x_c$ | 750                                                                | 919.9 | 1011   | 1091   | 1170   | 1274.9 | 1334.8 | 1696.4 |
| Nd2         | A     | 18.7                                                               | 71.7  | 21.7   | 47.3   | 31.9   | 44.4   | 82     | 174.5  |
|             | w     | 91                                                                 | 116.5 | 76     | 98.7   | 109.8  | 147    | 400.9  | 893    |
|             | $x_c$ | 751.3                                                              | 921.2 | 1012.4 | 1088   | 1163.6 | 1266.6 | 1333.3 | 1761.2 |
| Nd4         | A     | 8.7                                                                | 35.5  | 10.1   | 21.8   | 15.3   | 22.7   | 43.7   | 112.5  |
|             | w     | 91.2                                                               | 119   | 74.2   | 96.5   | 111.5  | 158.4  | 426.4  | 984    |
|             | $x_c$ | 749                                                                | 919.9 | 1005.4 | 1089.5 | 1243   | 1276.5 | 1278.4 | 1729.4 |
| Nd6         | A     | 12.8                                                               | 52.9  | 5      | 47     | 46.3   | 41.9   | 71.7   | 73     |
|             | w     | 83                                                                 | 118.7 | 53.7   | 141.3  | 199    | 543.6  | 522.2  | 740.7  |
|             | $x_c$ | 749.9                                                              | 922.2 | 1015.9 | 1086.5 | 1158   | 1253.6 | 1336.9 | 1744.9 |
| Nd8         | A     | 13.5                                                               | 61.8  | 13.6   | 31.7   | 20.7   | 41.5   | 78.3   | 141.7  |
|             | w     | 85.4                                                               | 130.6 | 72.7   | 91.9   | 106.2  | 178.3  | 444    | 912.3  |
|             | $x_c$ | 750.5                                                              | 924.2 | 1019.6 | 1087.9 | 1157.7 | 1252.4 | 1348   | 1827.5 |
|             | A     | 13.9                                                               | 65.4  | 13.8   | 30.6   | 21.5   | 42.6   | 88     | 162.2  |
|             | w     | 86.4                                                               | 139   | 73.7   | 89.4   | 107.5  | 179.5  | 457    | 883.4  |

**Table 2**

Composition, glass ionicity,  $I_c\%$ , longitudinal and shear ultrasonic wave velocities ( $V_L$ ,  $V_S$ ), elastic moduli ( $L$ ,  $G$ ,  $K$  and  $E$ ), Poisson's ratio ( $\sigma$ ), mean ultrasonic wave velocity ( $U_m$ ), Depye's temperature ( $\theta_D$ ), microhardness ( $H$ ), acoustic impedance ( $Z$ ), softening temperature ( $T_S$ ), Cauchy relation ( $G_{12}$ ), fugacity ( $f_g$ ), fractal bond connectivity ( $d$ ) and thermal expansion coefficient ( $\alpha_p$ ) of glasses.

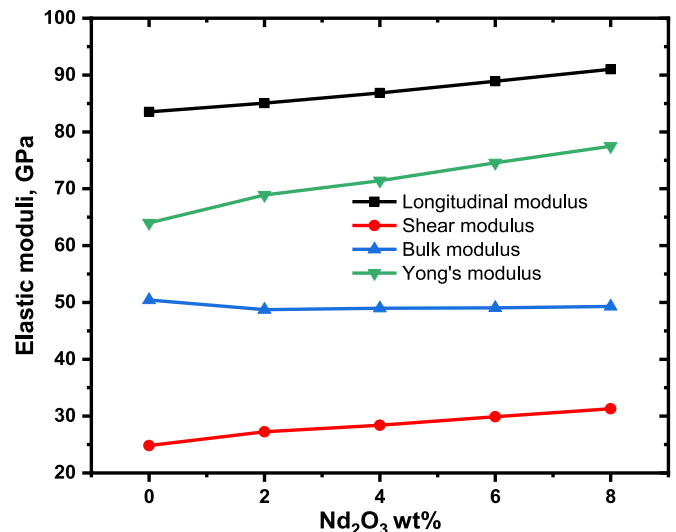
| Parameters                                             | Nd0     | Nd2     | Nd4     | Nd6     | Nd8     |
|--------------------------------------------------------|---------|---------|---------|---------|---------|
| $I_c\%$                                                | 74.57   | 74.72   | 74.87   | 75.02   | 75.16   |
| $V_L$ (m/s)                                            | 5145    | 5170    | 5197    | 5235    | 5280    |
| $V_S$ (m/s)                                            | 2805    | 2926    | 2972    | 3036    | 3096    |
| $L$ (GPa)                                              | 83.52   | 85.07   | 86.84   | 88.91   | 91.01   |
| $G$ (GPa)                                              | 24.82   | 27.25   | 28.40   | 29.90   | 31.29   |
| $\sigma$                                               | 0.289   | 0.264   | 0.257   | 0.247   | 0.238   |
| $K$ (GPa)                                              | 50.42   | 48.74   | 48.98   | 49.04   | 49.29   |
| $E$ (GPa)                                              | 63.97   | 68.90   | 71.40   | 74.56   | 77.48   |
| $U_m$ (m/s)                                            | 3128.61 | 3253.95 | 3302.21 | 3369.20 | 3432.38 |
| $\theta_D$ (K)                                         | 379.62  | 394.41  | 400.06  | 407.83  | 414.81  |
| $H$ (GPa)                                              | 3.50    | 4.28    | 4.60    | 5.05    | 5.47    |
| $Z \times 10^7$ (Kg.m <sup>-2</sup> .s <sup>-1</sup> ) | 1.62    | 1.65    | 1.67    | 1.70    | 1.72    |
| $T_S$ (K)                                              | 856.64  | 943.45  | 984.82  | 1039.46 | 1092.98 |
| $C_{12}$ (GPa)                                         | 33.87   | 30.57   | 30.04   | 29.10   | 28.43   |
| $G/C_{12}$                                             | 0.733   | 0.891   | 0.945   | 1.028   | 1.101   |
| $d$                                                    | 1.970   | 2.236   | 2.320   | 2.439   | 2.540   |
| $f_g$                                                  | 0.069   | 0.088   | 0.094   | 0.103   | 0.111   |
| $\alpha_p \times 10^3$ (K <sup>-1</sup> )              | 119.35  | 119.93  | 120.56  | 121.44  | 122.48  |

**Fig. 3.** Variation of longitudinal,  $V_L$ , and shear,  $V_S$ , ultrasonic velocities with  $Nd_2O_3$  (wt%) content.

$$\text{Mean sound velocity, } U_m = \left[ \frac{3V_L^3 V_S^3}{2V_L^3 + V_S^3} \right]^{1/3} \quad (8)$$

Fig. 4 depicts the elastic moduli ( $L$ ,  $G$ ,  $K$ , and  $E$ ) and their values are listed in Table 2. The elastic moduli increased as the amount of  $Nd^{3+}$  in the system increased. The  $E$  changed from 63.97 to 77.48 GPa; however, the  $G$  changed from 24.82 to 31.29 GPa. The  $K$  varied from 48.74 to 50.42 GPa, the  $L$  varied from 83.52 to 91.01 GPa and the values of  $\sigma$  varied from 0.289 to 0.238. The rise in elastic moduli, except for the  $K$ , might be attributed to network shrinking around the  $Nd^{3+}$  ions. The  $Nd^{3+}$  ions have a strong field that attracts their neighbors' atoms and increases the ion-dipole interaction, resulting in higher elastic moduli [35,36]. The velocity and elastic moduli of the glass increase as the rigidity of the glass increases. According to Bridge-Higazy's compression model, the shear modulus is proportional to the packing density [37,38], whereas the bulk modulus is proportional to the number of network bonds per unit volume [39]. This means that when the  $Nd_2O_3$  concentration rises, the network rigidity rises faster than the packing density.

The hardness and elastic characteristics of LZP:Nd glasses may be improved by adding  $Nd_2O_3$  content due to the rise in network connection and bonding strength. Covalent bonds are more difficult to break than ionic bonds, according to the relationship between glass network rigidity and bond strength, and hence the former might enhance network rigidity. With the addition of  $Nd_2O_3$ , more covalent Nd-O replace ionic Li-O bonds in the investigated LZP glasses [40], resulting in a more rigid glass network and therefore larger elastic moduli. In this

**Fig. 4.** Elastic moduli variation with  $Nd_2O_3$  (wt%) content in LZP glass system.

situation, the rise in covalency leads to increased micro-hardness. The dissociation energy for P-O and Nd-O bonds are 589 kJ/mol and 703 kJ/mol, respectively, indicating that when Nd<sub>2</sub>O<sub>3</sub> introduces to the glass, its network becomes more rigid. As the glass hardness and rigidity rise, the two velocities increase as well. So that, the addition of a greater dopant concentration of Nd<sup>3+</sup> ions causes the glass rigidity to rise. The ultrasonic velocities of glass have increased dramatically as a result of this. Furthermore, according to Shaaban et al. [41] and Veit and Russel [42], the rise in the elastic module is caused by packing linkage, coordination number, and density.

The Poisson effect, or deformation of a material in directions perpendicular to the specific direction of stress, is measured by Poisson's ratio ( $\sigma$ ). In general, a Poisson's ratio of 0.1 to 0.2 indicates a high cross-link density, whereas a Poisson's ratio of 0.3 to 0.5 indicates a low cross-link density. With the rise in Nd<sub>2</sub>O<sub>3</sub>,  $\sigma$  reduce from 0.289 to 0.238 in this work (see Table 2 and Fig. 5). The reduction in the  $\sigma$  value as Nd<sub>2</sub>O<sub>3</sub> concentration increases might be attributed to a rise in the glass ionicity induced by increased polar bonding in the modifiers [43]. In addition, the insertion of covalent links into the glass network caused by P-O-Nd bonds explains the reduction in the  $\sigma$ . As displayed in Fig. 5, the  $G/C_{12}$  ratio rose from 0.733 to 1.101 as the Nd<sub>2</sub>O<sub>3</sub> concentration increased. This ratio is used to determine the force field's characteristics. When the kind of force is central, ( $G/C_{12} = 1$ ) is seen, and when the type of force is non-central, ( $G/C_{12} \neq 1$ ).

The amount of lattice vibration in solid materials, as well as the excitation temperature of vibration in the materials, was investigated by Debye temperature. The softening temperature,  $T_s$ , term refers to the temperature change from viscous to plastic flow. Glasses' temperature stability was also determined by their softening temperatures. The variation of the Debye ( $\theta_D$ ) and softening temperatures ( $T_s$ ) with the concentration of Nd<sub>2</sub>O<sub>3</sub> are presented in Fig. 6 and Table 2. The ( $T_s$ ) and the ( $\theta_D$ ) were calculated using the following relations;

$$\text{Softening temperature, } T_s = \left( \frac{\text{Molecular weight of the glass}}{507.4^2 P} \right) V_s^2 \quad (9)$$

$$\text{Debye temperature, } \theta_D = \frac{h}{k_B} \left( \frac{3PN_A}{4\pi V_m} \right)^{1/3} U_m \quad (10)$$

where  $h$  is the Plank's constant,  $P$  is the number of atoms in the chemical formula,  $k_B$  is the Boltzmann constant,  $N_A$  is the Avogadro's number and  $V_m$  is the molar volume of the given sample. As the concentration of Nd<sub>2</sub>O<sub>3</sub> increased the values of the ( $T_s$ ) and the ( $\theta_D$ ) increased from 856.64 to 1092.98 K and 379.62 to 414.81 K, respectively. The rise in the  $T_s$  value with the increase of the Nd<sub>2</sub>O<sub>3</sub> content might be associated with an increase in the rigidity and connectivity within the glass matrix.

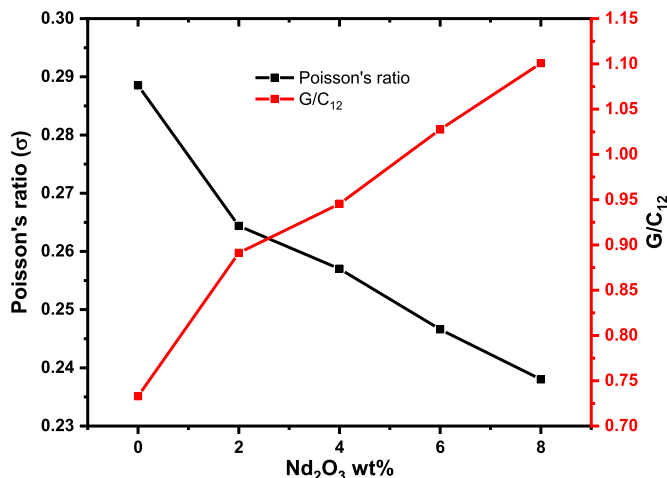


Fig. 5. Poisson's ratio and ( $G/C_{12}$ ) ratio vs. Nd<sub>2</sub>O<sub>3</sub> (wt%) for LZF glasses.

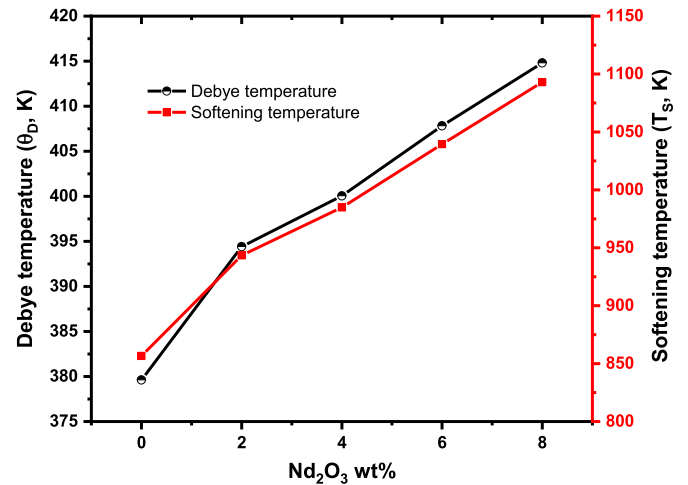


Fig. 6. Softening temperature and Debye temperature variation with Nd<sub>2</sub>O<sub>3</sub> (wt%) content in LZF glass system.

Fractal bond connectivity of a glass sample ( $d$ ) is a suitable parameter and provides information about the dimensionality of the glass arrangement. An additional significant parameter in the current study is fugacity  $f_g$  which is also called the fluctuation free volume in sample glasses, which describes the amount of rigidity of the synthesized glass system. The following relations are used to compute these parameters [44]:

$$\text{Fractal bond connectivity, } d = \frac{4G}{K} \quad (11)$$

$$\text{Fugacity, } \left[ f_g \ln \left( \frac{1}{f_g} \right) \right] = \frac{(1 - 2\sigma)^2}{2(1 + \sigma)} \quad (12)$$

The values of ( $d$ ) and ( $f_g$ ) are listed in Table 2 and displayed in Fig. 7. The two parameters have the same behavior and their values increase from 1.970 to 2.540 and from 0.069 to 0.111, respectively, with an increase in Nd<sub>2</sub>O<sub>3</sub> content. An increase in ( $d$ ) and ( $f_g$ ) of synthesized glasses is assumed to be due to an increase in the amount of NBOs and rigidity, respectively [44].

The micro-hardness ( $H$ ) of sample glasses is used to determine the amount of stress necessary to remove free volume. The variation of thermal expansion coefficient ( $\alpha_p$ ) and the microhardness ( $H$ ) with Nd<sub>2</sub>O<sub>3</sub> is displayed in Fig. 8 and Table 2. The ( $\alpha_p$ ) and ( $H$ ) were estimated using the following relations [45]:

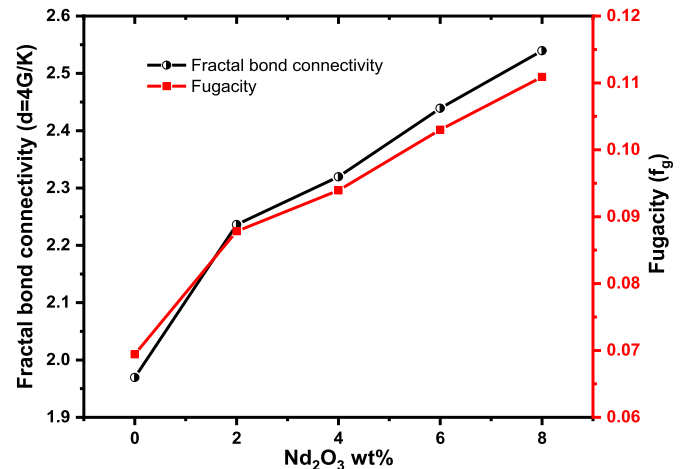


Fig. 7. Fractal bond connectivity and Fugacity variation with Nd<sub>2</sub>O<sub>3</sub> (wt%) content in LZF glass system.

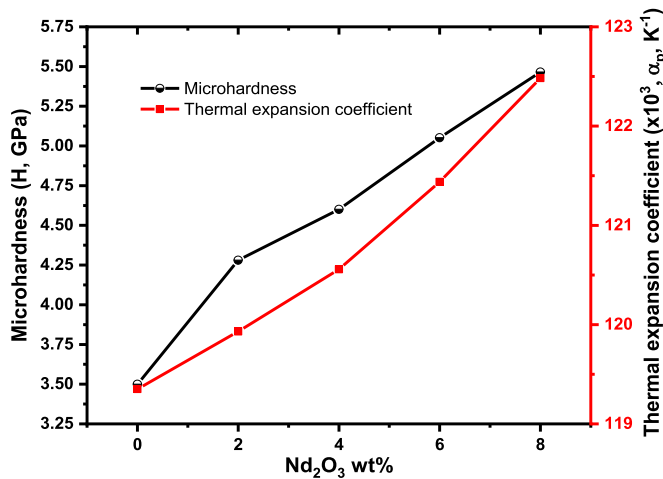


Fig. 8. Microhardness and thermal expansion coefficient variation with  $\text{Nd}_2\text{O}_3$  wt% content in LZP glass system.

$$\text{Thermal expansion coefficient, } \alpha_p = 23.2(V_L - 0.57457) \quad (13)$$

$$\text{Microhardness, } H = \frac{(E - 2\sigma E)}{(6 + 6\sigma)} \quad (14)$$

The ( $\alpha_p$ ) increases from  $119.35 \times 10^3 \text{ K}^{-1}$  to  $122.48 \times 10^3 \text{ K}^{-1}$  and the (H) increases from 3.500 GPa to 5.465 GPa. With an increase in the  $\text{Nd}_2\text{O}_3$  concentration, the ( $\alpha_p$ ) reflects a reduction in the thermal stability of the glasses, and the rise in their covalence leads to the increase in the (H).

### 3.4. Nuclear radiation shielding properties

The mass attenuation coefficient (MAC) of glasses prepared (Nd0-Nd8) using Phy-X/PSD software was obtained at gamma-ray energy in the range from 0.015 to 15 MeV [46]. Furthermore, XCOM database software was used to confirm the results. The results for the MAC values against the gamma-ray energy and the content of the neodymium oxide are shown in Fig. 9. Based on the results shown in Fig. 9 and Table 3, the MAC values for all composite glasses give the highest value at 0.015 MeV. Also, the highest values of MAC ranged between 15.294 and  $17.391 \text{ cm}^2 \text{ g}^{-1}$  for Nd0 and Nd8 glasses, respectively. After 0.015 MeV, the MAC values for all prepared glass showed a decrease due to the

photoelectric reaction (PE) predominant in this region [6].

Starting from 0.04 to 0.3 MeV the results showed that, the MAC values of the composite glasses showed a large variation in the MAC between prepared samples, this variation drop decreases as the proportion of  $\text{Nd}_2\text{O}_3$  increases. The results also indicated that the addition of  $\text{Nd}_2\text{O}_3$  improves the MAC of the prepared glasses in particular at this energy region and in general for all other energies. Above 0.3 MeV, the MAC values of all glass decrease slowly due to pair production (PC). This is because the pair production has a reaction cross-section proportional to  $\log(E)$ . The highest MAC values were for the Nd8 glass sample and ranged between 0.025 and  $29.548 \text{ cm}^2 \text{ g}^{-1}$ , while the lowest MAC values were for the Nd0 and increased from 0.024 to  $26.034 \text{ cm}^2 \text{ g}^{-1}$  for gamma-ray energy in the range from 0.015 to 15 MeV.

Fig. 10 shows the Linear attenuation coefficient (LAC) of prepared glasses. From the figure, it can be noted that the values of LAC are affected by the energy values of the gamma-ray. Also, it depends on the chemical composition of the glass. The highest LAC values were found at 0.015 MeV and the lowest at 15 MeV. These values were varied between 0.99 and  $1.024 \text{ cm}^{-1}$  and between 3 and  $7 \text{ cm}^{-1}$  for Nd0 and Nd8 glass samples, respectively. It was also observed that as the gamma-ray's energy increases, the LAC values decrease. The behavior of the LAC with increasing gamma-ray energy is similar to that of the MAC.

Figs. 11 and 12 show the changes of half-value layer (HVL) and mean free path (MFP) quantities with gamma photon energy in the range from 0.015 to 15 MeV, accordingly, all prepared glass samples Nd0-Nd8 show very small differences in HVL and MFP at energy from 0.015 to 0.04 MeV and it increases slowly. In the energy range from 0.04 to 0.6 MeV, the difference of HVL and MFP values for all presented glass samples is clear, the highest values were for the Nd0 sample, while the lowest values were for the Nd8 sample and the values decreased with increasing the amount of neodymium oxide. From energy 0.6 to 3 MeV, the values of HVL and MFP were close for all samples and increased rapidly with increasing energy. Above the energy 8 MeV, all HVL and MFP values increase rapidly to their maximum values, with clear differences between the values of all prepared glass samples. The lowest values of the HVL of the presented samples were at the energy of 0.015 MeV and it was about 0.0080, 0.0081, 0.0077, 0.0070 and 0.0070 cm, while the highest values were at the energy of 15 MeV and it was about 9.229, 9.069, 8.754, 8.544, and 8.367 cm. Also, the lowest MFP values were at energy 0.015 MeV about 0.0120, 0.0127, 0.0111, 0.0110, and 0.0100  $\text{cm}^{-1}$  for samples Nd0-Nd8, respectively. While the highest values were at energy 15 MeV about 13.314, 13.084, 12.630, 12.326, and 12.071  $\text{cm}^{-1}$  for samples Nd0-Nd8, respectively.

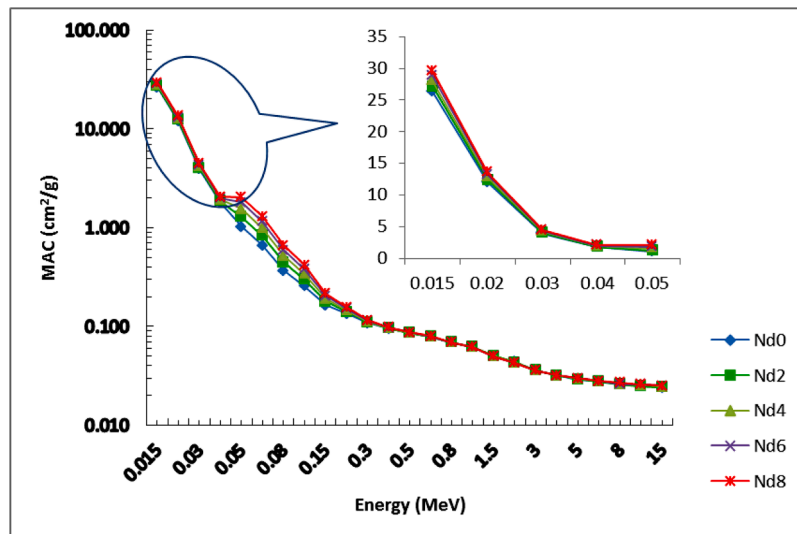
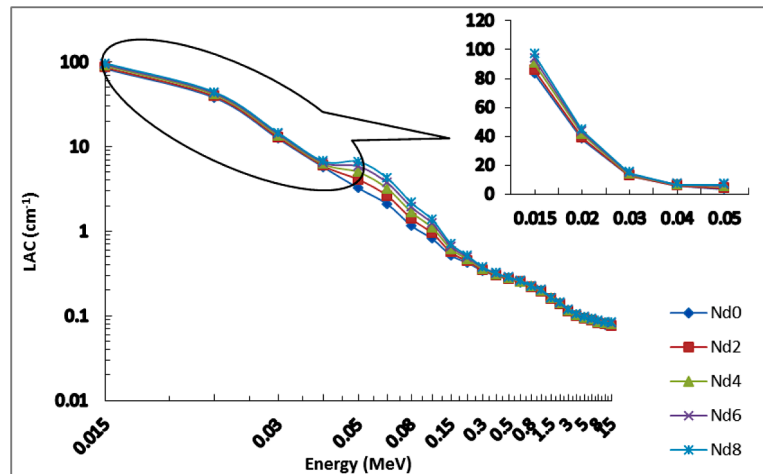


Fig. 9. Mass attenuation coefficient for presented glass samples.

**Table 3**

Obtained MAC from XCOM database and Phy-X/PSD program.

| Energy (MeV) | Nd0       |        | Nd2       |        | Nd4       |        | Nd6       |        | Nd8       |        |
|--------------|-----------|--------|-----------|--------|-----------|--------|-----------|--------|-----------|--------|
|              | Phy-X/PSD | XCOM   | Phy-X/PSD | XCOM   | Phy-X/PSD | XCOM   | Phy-X/PSD | XCOM   | Phy-X/PSD | XCOM   |
| 0.015        | 29.510    | 29.548 | 28.770    | 28.785 | 27.960    | 27.996 | 27.220    | 27.165 | 26.430    | 26.304 |
| 0.02         | 13.490    | 13.509 | 13.140    | 13.150 | 12.760    | 12.778 | 12.410    | 12.387 | 11.950    | 11.982 |
| 0.03         | 4.446     | 4.451  | 4.325     | 4.327  | 4.193     | 4.198  | 4.072     | 4.063  | 3.848     | 3.923  |
| 0.04         | 2.052     | 2.054  | 1.995     | 1.996  | 1.933     | 1.935  | 1.875     | 1.871  | 1.742     | 1.805  |
| 0.05         | 2.031     | 2.038  | 1.796     | 1.798  | 1.544     | 1.550  | 1.297     | 1.289  | 0.970     | 1.019  |
| 0.06         | 1.294     | 1.298  | 1.147     | 1.148  | 0.989     | 0.993  | 0.835     | 0.830  | 0.623     | 0.661  |
| 0.08         | 0.660     | 0.662  | 0.592     | 0.593  | 0.519     | 0.521  | 0.447     | 0.445  | 0.342     | 0.367  |
| 0.1          | 0.416     | 0.417  | 0.379     | 0.379  | 0.339     | 0.340  | 0.300     | 0.299  | 0.239     | 0.256  |
| 0.15         | 0.216     | 0.216  | 0.204     | 0.204  | 0.191     | 0.191  | 0.178     | 0.178  | 0.155     | 0.164  |
| 0.2          | 0.157     | 0.157  | 0.152     | 0.152  | 0.146     | 0.146  | 0.140     | 0.140  | 0.129     | 0.134  |
| 0.3          | 0.115     | 0.115  | 0.114     | 0.114  | 0.112     | 0.112  | 0.110     | 0.110  | 0.106     | 0.108  |
| 0.4          | 0.098     | 0.098  | 0.097     | 0.097  | 0.096     | 0.096  | 0.096     | 0.095  | 0.093     | 0.095  |
| 0.5          | 0.087     | 0.087  | 0.087     | 0.087  | 0.086     | 0.086  | 0.086     | 0.086  | 0.085     | 0.086  |
| 0.6          | 0.080     | 0.080  | 0.079     | 0.079  | 0.079     | 0.079  | 0.079     | 0.079  | 0.078     | 0.079  |
| 0.8          | 0.069     | 0.069  | 0.069     | 0.069  | 0.069     | 0.069  | 0.069     | 0.069  | 0.068     | 0.069  |
| 1            | 0.062     | 0.062  | 0.062     | 0.062  | 0.062     | 0.062  | 0.062     | 0.062  | 0.062     | 0.062  |
| 1.5          | 0.050     | 0.050  | 0.050     | 0.050  | 0.050     | 0.050  | 0.050     | 0.050  | 0.050     | 0.050  |
| 2            | 0.043     | 0.043  | 0.043     | 0.043  | 0.044     | 0.043  | 0.044     | 0.044  | 0.043     | 0.044  |
| 3            | 0.036     | 0.036  | 0.036     | 0.036  | 0.036     | 0.036  | 0.036     | 0.036  | 0.036     | 0.036  |
| 4            | 0.032     | 0.032  | 0.032     | 0.032  | 0.032     | 0.032  | 0.032     | 0.032  | 0.032     | 0.032  |
| 5            | 0.030     | 0.030  | 0.030     | 0.030  | 0.030     | 0.030  | 0.029     | 0.029  | 0.029     | 0.029  |
| 6            | 0.028     | 0.028  | 0.028     | 0.028  | 0.028     | 0.028  | 0.028     | 0.028  | 0.028     | 0.028  |
| 8            | 0.027     | 0.027  | 0.026     | 0.026  | 0.026     | 0.026  | 0.026     | 0.026  | 0.026     | 0.026  |
| 10           | 0.026     | 0.026  | 0.026     | 0.026  | 0.025     | 0.025  | 0.025     | 0.025  | 0.025     | 0.025  |
| 15           | 0.025     | 0.025  | 0.025     | 0.025  | 0.025     | 0.025  | 0.024     | 0.024  | 0.024     | 0.024  |

**Fig. 10.** Linear attenuation coefficient (LAC) for prepared glass samples.

For the practical use of glass as gamma-ray shields, It is appropriate to compare the HVL and the MAC of prepared glass with some standard shielding concrete as well as commercial glass types [47]. The shorter the HVL of the material means that the interaction of photons is largest and therefore, the shielding properties of the material are better. The prepared glass sample Nd8, which gave the best results for HVL and MAC, as compared with ordinary concrete (OC), Barite concrete (Ba), RS 360 and RS 253 G18 glasses, and the results are shown in Figs. 13 and 14, respectively. From the obtained results, it can be noticed that the HVL of Nd8 present glass is higher than Ba and RS 360, whereas less than OC and RS 253 G18. Also, from Fig. 14, the results show that the MAC for the presented sample Nd8 gave the best results compared to the OC, as well as RS 253 G18 glass, while the MAC of sample Nd8 was lower than the MAC of Ba concrete and RS 360 glass.

The fast neutron removal cross-section ( $\Sigma_R$ ) also was obtained for the prepared samples and the results are shown in Fig. 15. The results show that the  $\Sigma_R$  values decreased with the increase in density and with the addition of  $\text{Nd}_2\text{O}_3$ . The  $\Sigma_R$  values for Nd0-Nd8 glasses were 0.1015

$\text{cm}^{-1}$ , 0.1025  $\text{cm}^{-1}$ , 0.1024  $\text{cm}^{-1}$ , 0.1024  $\text{cm}^{-1}$  and 0.1020  $\text{cm}^{-1}$ , respectively. It can be noticed that the Nd0 glass system was more effective in capturing neutrons compared with the other samples. It is also clear that the prepared glass systems are more effective in reducing neutrons when compared to water and paraffin ( $\Sigma_R = 0.101 \text{ cm}^{-1}$ ), which are conventional neutron mediators.

#### 4. Conclusions

The following is a summary of the main conclusions:

- Based on the FTIR analyses, it can be deduced that both the symmetric vibrations of P-O-P and/or Zn-O-Zn groups and the vibrations of  $\text{P-O}^-$  groups are displayed a reverse behavior for the variations of their wavenumber positions with  $\text{Nd}_2\text{O}_3$  content.
- As the  $\text{Nd}_2\text{O}_3$  concentration rose, the sound velocity and elastic characteristics including Young's modulus, shear modulus, and longitudinal modulus increased, but Poisson's ratio reduced. The

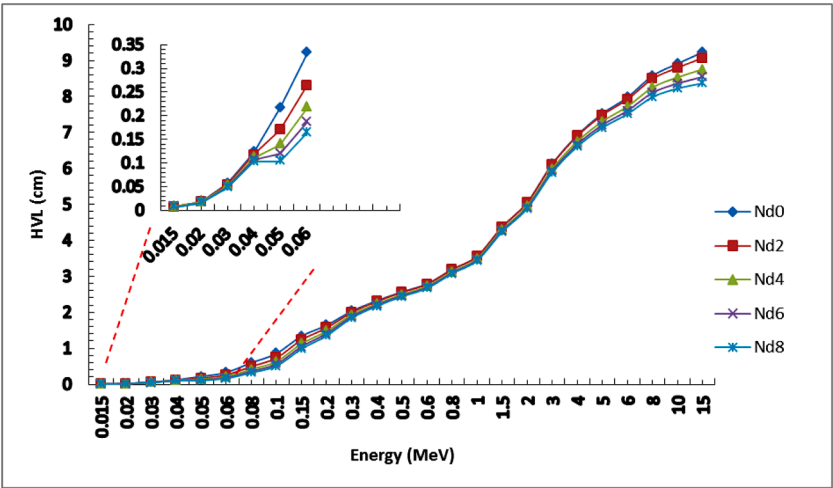


Fig. 11. Half-value layer (HVL) for prepared glass samples.

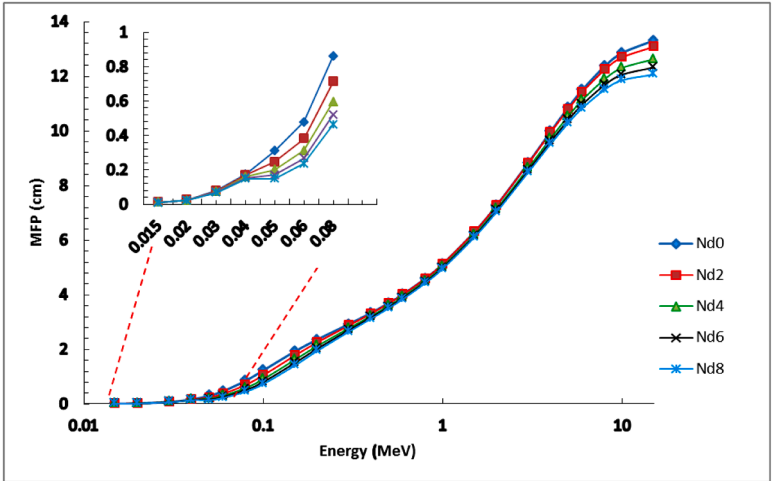


Fig. 12. Mean free path (MFP) for prepared glass samples.

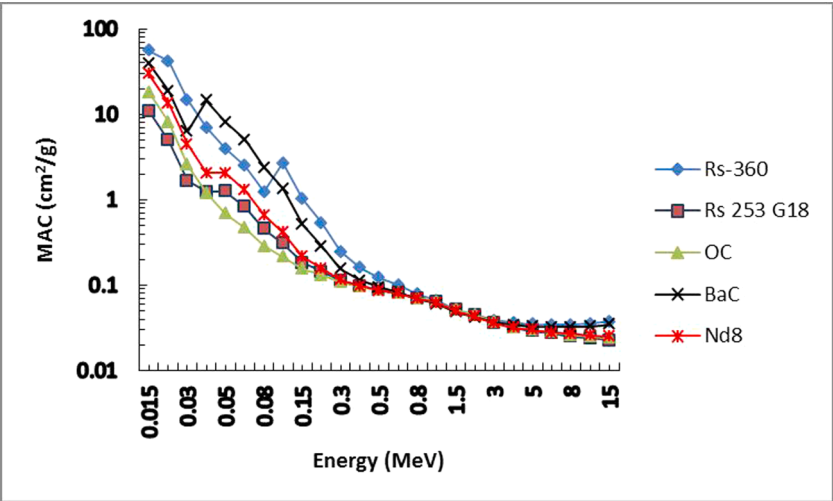


Fig. 13. Comparison between MAC for Nd8 present sample and commercial materials.

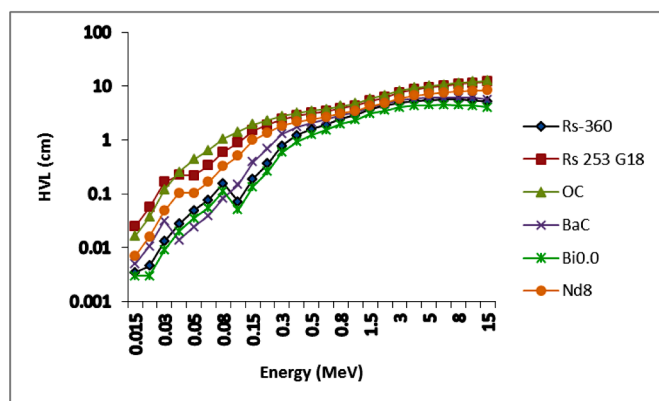


Fig. 14. Comparison between HVL for Nd8 present sample and commercial materials.

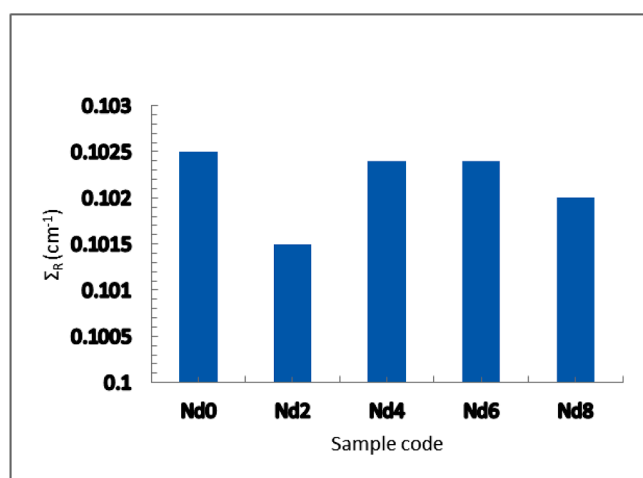


Fig. 15. The fast neutron removal cross-section ( $\Sigma_R$ ) for prepared glass samples.

reduction in Poisson's ratio indicates that the glasses' rigidity has risen. As a result of the molar mass of neodymium oxide compared to the other components in LZP glass leading to an increase in the density, ultrasonic velocities, and elastic moduli. The increase in the elastic moduli is also attributed to the local constriction around the neodymium ions that have a high field strength and may be attributable to the increase in rigidity and network structure connectivity.

- The introduction of  $\text{Nd}_2\text{O}_3$  increases the softening and Debye temperatures of the glass system due to an increasing number of atoms.
- The fractal bond connectivity plays a key role in determining changes in the structural and acoustical characteristics of the studied glasses.
- The ionic nature, I%, of the LZP:Nd glasses was detected from obtained values of ionic characteristic factor.
- The addition of  $\text{Nd}_2\text{O}_3$  improves the attenuation properties of gamma-rays in the range from 0.015 to 15 MeV. Also, the addition of  $\text{Nd}_2\text{O}_3$  reduces (HVL), as well as (MFP). Conversely, the addition of  $\text{Nd}_2\text{O}_3$  reduces  $\Sigma_R$ . Compared with ordinary concrete and commercial glass used for gamma-rays shielding, the prepared glass samples were better than ordinary concrete and RS 360 glass.
- The values of elastic and nuclear radiation shielding parameters of the glass system were found to be sensitive to the addition of  $\text{Nd}_2\text{O}_3$ .

#### CRediT authorship contribution statement

Mohammed A. Algradee: Conceptualization, Writing – original

draft, Methodology, Investigation, Visualization, Formal analysis, Writing – original draft. Emran Eisa Saleh: Software, Investigation, Conceptualization, Formal analysis, Writing – original draft. O.M. Samir: Data curation, Writing – review & editing. A.B. Alwany: Data curation, Writing – review & editing. Tharwat M. EL Sherbini: Writing – review & editing, Supervision, Project administration, Validation.

#### Declaration of Competing Interest

None.

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